

INTEGRATED FACILITY OPERATIONS PLATFORM

Full System Build Specification

Payment • Management • Operations • Booking • Inventory • Staff • Reporting

Prepared for the Otueke Property Project

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Version 1.0

1. Executive System Definition

The Integrated Facility Operations Platform is the central digital operating system for the property. It connects payment, facility operations, hospitality service, inventory, bookings, ticketing, staff identity, attendance, management reporting, remote access and customer-facing online services into one controlled platform.

The system is designed as a hybrid local-first and cloud-connected environment. Critical on-site operations continue through the local server and property network, while cloud services provide secure remote management, public website access, online booking, backup and synchronization.

1.1 Core design principles

- One property-wide system rather than disconnected applications for each business unit.
- Every operational area is configured as a Facility Unit with capabilities enabled as required.
- Payments, orders, bookings, tickets, inventory and staff actions are auditable.
- Critical property operations must not depend entirely on internet availability.
- Remote management must not require exposing the physical property server directly to the public internet.
- Roles and permissions control what each staff member can view or perform.
- Architecture must accommodate additional facilities and a future hotel without rebuilding the Phase 1 platform.

1.2 Phase 1 facility scope

Facility / Area	Primary System Functions
Main Reception	Payments, pool/sports sales, sports bookings, equipment rental, ticket issue, customer service
Restaurant	Hospitality POS, waiter orders, KDS routing, receipts, stock consumption
Indoor Club	Open tabs, waiter service, dispensing workflow, settlement at exit
Beauty Spa	Service POS, appointments, bookings, customer records
Pool Area / Pool Bar	Pool ticketing/validation; Pool Bar order workflow; payments routed through Reception
Cafe / Cyber Cafe	POS and service transactions
Main Kitchen	KDS production workflow
Sports Arena	Football, lawn tennis and basketball booking/validation
Bush Bar / Event Centre	Hospitality POS, KDS/dispensing, event-related operations
Salon	Service POS
Supermarket / Super Store	Retail POS, barcode scanning, stock
Sports Store	QR entitlement validation, equipment release/return; payment at Reception
Main Store	Central inventory and transfers
Accounts / Manager / IT	Finance, management, administration, configuration and support

1.3 Explicit Phase 1 exclusion

The future hotel is not built and is excluded from the current Phase 1 scope and price. The platform will remain expansion-ready for a future Property Management System (PMS), room inventory, reservations, housekeeping, room access cards, room charges and related hotel functions.

2. Configurable Facility Architecture

The software must not hardcode the property around today's facility names. Each operational area is represented as a configurable Facility Unit.

Hierarchy	Purpose
Property	Top-level organization/site
Facility Unit	Restaurant, Spa, Pool, Sports, Club, Supermarket, etc.
Operating Point / Resource	Counter, court, room, table group, service point, booking resource
Terminal / Device	POS, KDS, tablet, scanner, NFC reader, printer
Product / Service	Food, drinks, retail items, tickets, rentals, appointments
Operating Rules	Payment timing, validation mode, capacity, permissions, stock, booking rules

2.1 Facility capabilities

- POS and receipts
- Ticketing and QR validation
- Bookings, time slots and capacity control
- Inventory and stock consumption
- Table service and open tabs
- Kitchen/bar routing
- Memberships and subscriptions
- Appointment scheduling
- Staff assignment and authorization
- Barcode/QR scanning
- Customer-facing online availability

3. Payment & POS System

Payment and sale/order records are separate entities. This supports split payments, open tabs, partial settlement, refunds, reconciliation and different payment timing rules across facilities.

Location	Fixed POS	Payment responsibility
Main Reception	2	Pool, Sports Arena/bookings, Sports Store purchases, equipment rental, Pool Bar transactions
Restaurant	1	Restaurant transactions
Indoor Club	1	Club tabs and settlement
Beauty Spa	1	Spa services
Bush Bar / Event Centre	1	Bush Bar/event transactions
Cafe / Cyber Cafe	1	Cafe/cyber transactions
Salon	1	Salon services
Supermarket	2	Retail checkout
TOTAL	10	Fixed payment terminals

3.1 Payment controls

- Payment timing is configurable per Facility Unit; the platform must not force payment-before-production everywhere.
- Indoor Club supports open tabs: orders can route to preparation/dispensing before final settlement.
- Sensitive actions such as voids, refunds, price overrides, discounts, complimentary items and stock adjustments require authorized supervisor rights.
- Each transaction records facility, operating point, terminal, staff actor, payment method, amount, time and relevant approval.
- Finance reconciliation must map internal transaction -> payment provider/reference -> settlement/bank record.
- Money values use fixed decimal types; financial records are not silently overwritten.

4. Hospitality Orders, KDS & Service Workflow

The hospitality workflow controls order creation, routing, preparation, dispensing, service and settlement. No controlled food or drink item should be dispensed outside the recorded workflow.

KDS Station	Purpose
Main Kitchen	Food production queue
Restaurant Counter	Restaurant order/status coordination
Pool Bar	Drink/dispensing queue
Bush Bar	Drink/dispensing queue

- Orders may originate from fixed POS terminals or authenticated waiter/supervisor tablets.
- Products are mapped to their production/dispensing destination.
- System records who created, accepted, prepared/dispensed, marked ready and served/released an order.
- Wait staff can create orders; restricted actions require supervisor approval.
- Real-time updates are delivered to relevant KDS/order screens.
- The smaller kitchen has no dedicated KDS in Phase 1 but the architecture and network remain expansion-ready.

5. Mobile Tablet Operations

Tablet Pool	Qty	Use
Shared waiter pool	12	Signed out from Reception per shift and assigned to staff/facility
Dedicated supervisor tablets	4	Restaurant, Indoor Club, Pool Bar, Bush Bar/Event Centre
Sports Entrance	1	QR booking/ticket validation
Sports Store	1	QR entitlement/equipment release and return
TOTAL	18	Android tablets

Tablet control records device ID, staff member, assigned facility, shift, checkout time, return time and device status. Sports tablets connect by Wi-Fi to the local system and do not require dedicated Ethernet or UPS.

6. Sports Booking, Ticketing & Equipment

Sports transactions are initiated and paid at Main Reception. The Sports Arena and Sports Store do not have payment terminals.

- Reception creates booking/ticket and accepts payment.
- Optional equipment rental and Sports Store items can be included in the same transaction.
- Customer receives receipt and QR entitlement.
- Sports Entrance scans the QR and validates status such as VALID, USED, EXPIRED or WRONG FACILITY.
- Sports Store scans the same entitlement and displays exactly what was paid/rented.
- Attendant records equipment released; rental items can later be marked returned.
- Entitlements prevent unauthorized double redemption.
- Booking can support whole-resource booking, individual capacity/slot booking, or both.

7. Ticketing, Booking & Membership Engine

7.1 Ticketing

- Per-person ticket issue is supported; e.g. 3 children + 2 adults can produce 5 unique tickets.
- Ticket output can be individual, combined, booking-based, digital, physical or both.
- Validation modes: none, single-use, multiple-entry, time-limited, entry/exit or staff approval.
- Pool is expected to use ticketing and single-use validation by default; rules remain configurable.

7.2 Bookings and appointments

- Sports resources can be booked by date, time slot, capacity or whole-resource rule.
- Spa and other service units can enable appointment scheduling.
- Availability is shared between Reception and the online booking portal to reduce double booking.
- Cancellation, rescheduling, capacity, blackout and staff/resource rules are configurable.

7.3 Memberships and subscriptions

Configurable field	Examples
Coverage	One facility, selected facilities or property-wide
Commercial rules	Price, duration, start/expiry, renewal, discounts
Usage	Visit limits, unlimited access, booking privileges, guest allowance
Lifecycle	ACTIVE, EXPIRED, SUSPENDED, CANCELLED, PENDING_RENEWAL
Validation	NFC, QR or membership ID

8. Inventory & Store Management

The Main Store is the central inventory point. Facility/sub-store inventory is managed from the workstation or POS already assigned to that facility; separate sub-store PCs are not required.

- Purchase/receipt into Main Store.
- Stock transfer from Main Store to unit/facility stock.
- Sale or consumption against unit stock.
- Wastage, returns and stock adjustments.
- Physical stock counts and variance reporting.
- Rental equipment issue/return where applicable.
- Recipe/BOM-based automatic ingredient consumption can be introduced as a later enhancement.

All stock movements must retain source, destination, item, quantity, staff actor, time and reason/reference.

9. Staff Identity, Roles & Attendance

Staff identity combines printed NFC PVC cards with system credentials. Sensitive fixed-station authentication uses NFC card plus PIN/password rather than NFC alone.

Fixed NFC reader allocation	Count
10 POS stations	10
Main Store workstation	1
Accounts workstation	1
Manager/Admin workstation	1
IT/Admin workstation	1
4 KDS stations	4
Active fixed readers	18
Spares	2
Procurement total	20

9.1 Roles

Initial roles include wait staff, bartender/dispenser, kitchen staff, cashier, storekeeper, unit supervisor, procurement, accountant, manager, IT/system admin and owner/super admin. Roles are permission sets, not dedicated computers.

9.2 Attendance

A network-connected biometric attendance terminal at the main gate records staff clock-in/clock-out and integrates with the central system where technically supported.

10. Management, Finance & Reporting

Management reporting follows Property -> Facility -> Operating Point -> Terminal -> Staff -> Transaction, allowing management to drill from high-level performance into individual operational events.

- Owner dashboard: property-wide sales, payments, bookings, stock indicators, attendance and operational status.
- Facility performance: revenue, transactions, orders, bookings and service activity by unit.
- Cashier/terminal reports and shift reconciliation.
- Payment provider and bank settlement reconciliation.
- Inventory balances, movement, wastage, adjustments and variance.
- Staff attendance and action audit.
- Ticket sales and validation reports.
- Sports bookings, utilization and equipment rental status.
- Membership status, renewals and usage.
- Remote access for authorized Owner, Manager/Admin and Accounts users.

11. Public Website & Online Booking Portal

The public-facing website and booking portal form a connected customer layer over the operational platform.

- Responsive official website for mobile, tablet and desktop.
- Property, facility, service, contact and location information.
- Sports booking, spa appointments, pool tickets and other enabled facility bookings.
- Online payment for eligible services.
- Booking confirmation and QR ticket/entitlement generation where applicable.
- Customer booking history and approved cancellation/rescheduling rules.
- Online availability synchronized with internal Reception operations.
- Expandable structure for future facilities and the separately commissioned hotel.

12. Local-First & Cloud Architecture

The property must continue to operate if public internet connectivity is temporarily unavailable. Terminals normally communicate with the local application server over the property LAN; cloud services provide remote access, public booking, backup, reporting and synchronization.

Layer	Responsibility
Local Windows Application Server	Primary operational API/services and local MySQL database
Property LAN / Wi-Fi	POS, KDS, workstations, tablets, printers and scanners
Cloud Environment	Remote management, website/booking, backup, synchronization and disaster recovery
Client Devices	Web/desktop interfaces, Android tablets, KDS and admin interfaces

12.1 Synchronization

- Synchronization-sensitive records use UUID identifiers.
- Sync states: LOCAL, QUEUED, SYNCING, SYNCED, CONFLICT and FAILED.
- Financial transactions are treated as immutable events with correction/reversal records rather than destructive edits.
- Idempotency controls prevent duplicate payment/order processing.
- Conflict handling is explicit for records that may be changed both locally and online.
- Optional emergency terminal queue can hold encrypted transactions during temporary loss of local-server connectivity.

13. Application & Technical Build Specification

Component	Selected technology / approach
Core API / Business Engine	C# / ASP.NET Core
Client applications	Native C#/.NET Windows POS; Flutter/Dart Android app; PHP Admin/Reporting and Booking Web; lightweight KDS client
Database	MySQL 8.4 LTS, InnoDB
Architecture	Modular monolith with clearly separated business modules
Real-time operations	SignalR / WebSocket-based updates for KDS and live operational screens
Local services	Windows/.NET worker services for device/hardware integrations where required
Caching / transient coordination	Redis where operationally justified
Local server OS	Windows
Source control / CI	Git-based source control and controlled deployment pipeline
Cloud	Suitable managed/cloud environment based on final hosting budget and operational requirements

13.1 MySQL requirements

- MySQL 8.4 LTS with InnoDB tables.
- Foreign-key constraints and deliberate indexing strategy.
- DECIMAL types for money; FLOAT/DOUBLE must not be used for financial amounts.
- UTC timestamps internally with localized presentation.
- UUIDs for synchronization-sensitive entities.
- Transactions around payment, stock and other multi-record integrity operations.
- Audit records for sensitive changes.
- Soft deletion only where appropriate; financial/audit history must remain traceable.
- Automated backups, restore testing and retention policy.

14. Security & Audit Requirements

- Role-based access control with least-privilege permissions.
- Individual staff accounts; no shared administrative credentials.
- NFC + PIN/password for sensitive fixed-station staff authentication.
- Multi-factor authentication for privileged remote Owner/Admin access where supported.
- HTTPS/TLS for online services.
- Guest Wi-Fi isolated from server, POS, CCTV and operational devices.
- Audit log captures actor, facility, terminal/device, action, timestamp, old value, new value and supervisor approval where applicable.
- Passwords stored using secure one-way password hashing.
- Session timeout and device/session revocation controls.
- Backup data protected from ordinary staff access.
- Public cloud services must not expose the local server/database directly to the internet.

15. Network & Infrastructure Integration

The operational network is separate from the previously installed CCTV cabling. New structured cabling serves Wi-Fi access points, Ethernet network points and operational devices.

Item	Qty	Purpose
CAT6 cable cartons	12	General operational Ethernet and AP cabling
25mm black conduit pipe	12	Protected cable routes
Network fittings/accessories	1 lot	Sockets, termination and related fittings
Installation, termination & testing	1 lot	Structured network deployment

Wi-Fi 6 access points	7	Operational wireless coverage
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Network design should logically separate SERVER, POS/OPERATIONS, CCTV, STAFF and GUEST traffic. Existing managed-switch/rack capacity is reused where sufficient; an additional managed switch is not included unless final port count proves it necessary.

16. Hardware & Device Baseline

Equipment	Qty	Specification / allocation
Touchscreen POS terminals	10	Windows commercial touchscreen POS/AIO; 2 Reception, 1 each Restaurant/Club/Spa/Bush Bar/Cafe/Salon, 2 Supermarket
80mm receipt printers	11	10 active + 1 spare
NFC readers	20	18 active fixed stations + 2 spare
Android tablets	18	REDMI Pad 2 8GB/256GB class; 12 waiter, 4 supervisor, 2 sports
Charging cabinet	1	Lockable 20-device charging/storage cabinet
KDS stations	4	Main Kitchen, Restaurant Counter, Pool Bar, Bush Bar
Wireless 2D QR scanners	2	Sports Entrance and Sports Store
Biometric attendance terminal	1	Main gate
ID card printer	1	Epson L8050 A4 ID Card & Photo Printer with ID-card printing tray/kit
Local application server	1	Business/server-class Windows server, 32GB ECC minimum, SSD/RAID storage
Backup NAS	1	NAS with redundant NAS-grade storage
Fixed workstations	4	Core i5-class, 16GB RAM, 512GB SSD, Windows 11 Pro; Main Store, Accounts, Manager/Admin, IT
Business laptops	2	Core i5 12th Gen+ class, 16GB RAM, 512GB NVMe SSD, FHD, Windows 11 Pro; Accounts/Manager remote work
Wi-Fi 6 access points	7	TP-Link Omada EAP650 class or equivalent
1000VA UPS	18	10 POS + 4 workstations + 4 KDS
Server/core online UPS	1	3kVA online UPS class
Supermarket 2D scanners	3	2 active + 1 spare
Tablet cases/accessories	18	Rugged case + screen protection
Server-room AC	1	1.5HP inverter split AC
Server-room furniture	1 set	Workstation desk, chair and sofa

17. Operational Workflows

17.1 Restaurant / Bar

Staff login -> create/select customer/table/tab -> add items -> route to KDS/dispensing -> preparation/ready status -> serve/release -> payment according to facility rule -> receipt -> stock and financial records update.

17.2 Sports

Reception selects resource/slot -> adds ticket/rental/store entitlement -> receives payment -> issues QR -> Sports Entrance validates access -> Sports Store validates paid equipment entitlement -> release/return recorded.

17.3 Inventory

Receive stock at Main Store -> approve/record transfer -> unit receives stock -> sale/consumption reduces unit stock -> wastage/returns/adjustments recorded -> management reviews balances and variances.

17.4 Staff

Staff clocks in at main gate -> signs into assigned workstation/tablet -> performs permitted actions -> supervisor authorizes restricted actions -> staff/device actions remain auditable -> clocks out and shared tablet is returned.

18. Core Data Domains

Domain	Principal records
Organization	Property, site, facility, operating point, resources
Identity	Users, staff, roles, permissions, NFC credentials, attendance
Catalog	Products, services, categories, prices, taxes/charges, facility availability
Hospitality	Orders, items, tabs, tables, KDS routing/status
Payments	Transactions, tenders, references, refunds/reversals, settlements
Inventory	Items, locations, receipts, transfers, movements, counts, wastage
Booking	Resources, slots, reservations, appointments, capacity
Ticketing	Tickets, QR entitlements, validation events
Membership	Plans, members, subscriptions, usage, renewal status
Devices	Terminals, tablets, printers, scanners, device assignments
Audit	User actions, approvals, old/new values, security events
Sync	Sync state, conflict state, idempotency and replication metadata

19. Deployment & Commissioning

- Confirm final facility map, operating points and network outlets.
- Install/terminate/test structured operational network and APs.
- Prepare server room, power protection, cooling, rack/network and local server.
- Install local application server, MySQL database, backup storage and services.
- Configure facilities, products/services, roles, permissions, ticket/booking rules and inventory locations.
- Deploy POS, workstations, KDS, printers, NFC readers, scanners and tablets.
- Configure staff identities, NFC cards and attendance integration.
- Configure cloud environment, website, remote management and online booking.
- Execute data integrity, payment, booking, ticket, inventory, offline and recovery tests.
- Train administrators, supervisors, cashiers, wait staff, storekeepers and relevant operational teams.
- Pilot selected facilities before full property go-live.
- Complete acceptance testing, documentation, handover and support transition.

20. Acceptance Criteria

- All configured facilities can transact according to their approved operating rules.
- Reception can process pool/sports/sports-store/pool-bar payments as designed.
- Restaurant/club/bar orders route correctly to KDS/dispensing destinations.
- Sports QR cannot be improperly redeemed twice and equipment issue/return is traceable.
- Inventory movements reconcile from Main Store to unit sale/consumption.
- NFC/PIN permissions and supervisor approvals function correctly.
- Critical local operations continue during simulated internet loss.
- Remote authorized users can access permitted management functions securely.
- Online bookings reflect in internal availability and do not create uncontrolled double booking.
- Backups can be restored in a controlled test.
- Reports reconcile to underlying transactions and audit records.

21. Future Hotel Expansion

When the hotel is built, a separately scoped PMS module can extend the same property platform with room inventory and rates, reservations, check-in/check-out, guest profiles, room assignment/change, deposits/balances, housekeeping, maintenance blocks, smart NFC/RFID room cards, card activation/expiry/lost-card cancellation, staff/master cards, door audit, room charges from restaurant/bar/spa and consolidated guest billing. Smart-lock/encoder hardware integration will be selected at that stage.

22. Scope Governance

This specification defines the intended Phase 1 functional and technical baseline. Final hardware brands, payment-provider integrations, third-party APIs, tax/accounting requirements, notification channels and any regulatory integrations should be confirmed during implementation. Material additions beyond this specification should be handled through controlled change requests so that cost, delivery and operational impact remain explicit.

23. Application Architecture - Authoritative Build Structure

This section supersedes any earlier implication that all user interfaces must use one web technology. The platform is intentionally split into lightweight operational clients and a centralized business engine. Business rules live in the API; POS, mobile, KDS and reporting clients consume the API and must not independently reimplement financial, inventory, ticketing or booking rules.

23.1 Core API and business engine

- ASP.NET Core / C# is the authoritative business API and service layer.
- MySQL 8.4 LTS with InnoDB is the operational relational database.
- The API owns orders, payments, refunds/reversals, inventory movements, bookings, tickets, memberships, permissions, synchronization and audit rules.
- Operational writes from all applications must pass through authenticated API endpoints rather than writing directly to MySQL.
- Real-time operational updates use SignalR/WebSockets where required.
- Local-first operation remains centered on the on-site Windows application server, with controlled cloud synchronization and remote services.

23.2 Lightweight Windows desktop POS

- Build one lightweight native Windows POS application in C#/.NET for fixed payment and operational terminals.
- The same application changes its screens and permissions according to Facility Unit, Operating Point, device registration and signed-in role.
- Reception, Restaurant, Indoor Club, Spa, Bush Bar, Cafe, Salon and Supermarket therefore use one configurable POS codebase rather than separate applications.
- POS responsibilities include authentication/NFC, catalog/service selection, cart/order/tab handling, payment capture workflow, receipt printing, barcode/QR/NFC device interaction and status display.
- The POS may keep an encrypted local cache/queue for resilience, but each terminal must not run an independent MySQL database.

23.3 Mobile attendant application

- Build one Flutter/Dart Android application for the 18 operational tablets.
- The 12 moving-attendant tablets use role- and facility-driven screens for waiter/order operations and shift/device assignment.
- The 4 supervisor tablets expose normal operational functions plus authorized supervisor actions.
- The Sports Entrance mode performs QR access validation; the Sports Store mode validates paid/rented entitlements and records release/return.
- The application should be lightweight, API-driven and capable of controlled caching/queueing for temporary connectivity interruptions.

23.4 KDS client

- Build a lightweight KDS client for Main Kitchen, Restaurant Counter, Pool Bar and Bush Bar.
- KDS receives routed orders in real time and supports accepted/in-progress/ready/dispensed status transitions according to permission.
- KDS does not own order pricing, payment or inventory business logic; those remain in the API.

23.5 Web administration and reporting

- Build the Owner/Management/Accounts/Administration web portal in PHP using a structured modern framework and modular architecture.

- The PHP portal is responsible for dashboards, reporting, configuration screens, finance/reconciliation views, staff/role administration, inventory oversight, bookings, memberships and operational management.
- Operational mutations must call the ASP.NET Core API. The PHP application must not duplicate payment, inventory, ticket, membership or booking business rules.
- Reporting may use purpose-built read endpoints or an approved read-only reporting connection/view where performance requires it.
- Privileged remote access requires HTTPS, role controls, auditability and MFA where supported.

23.6 Public website and booking portal

- The public website and online booking portal may be implemented in PHP/web technology and must consume the same authoritative API.
- It supports facility discovery, sports booking, spa appointments, pool/ticketed services, eligible online payments, QR confirmations, booking history and approved cancellation/rescheduling rules.
- Internal and online availability must share the same booking rules to prevent uncontrolled double booking.

23.7 Source-code repository structure

Repository	Primary stack	Responsibility
otueke-api	ASP.NET Core / C#	Core API, business rules, local/cloud services, integrations
otueke-pos-desktop	C# / .NET	Lightweight native Windows POS
otueke-mobile	Flutter / Dart	Moving attendants, supervisors, sports validation/store modes
otueke-admin-web	PHP	Owner, Accounts, Manager, reporting and configuration portal
otueke-booking-web	PHP / Web	Public website and customer booking portal
otueke-kds	Lightweight web/.NET client	Kitchen/bar production and dispensing display
otueke-infrastructure	Configuration / scripts	Deployment templates, environment examples, migrations/runbooks without secrets
otueke-docs	Markdown / diagrams	Architecture decisions, API contracts, workflows and operational documentation

23.8 Repository and engineering rules

- All repositories are private under the GitHub account/organization prinzderick.
- No production secrets, API keys, database passwords, payment credentials or private certificates are committed to Git.
- Each repository contains README, setup instructions, environment example, branching guidance, tests and CI checks appropriate to the stack.
- API contracts are versioned and documented so desktop, Flutter, PHP and KDS clients can evolve independently without duplicating business logic.
- Database schema/migrations have one authoritative owner in the API/backend repository; other applications do not maintain competing schema migrations.
- Use feature branches and pull requests for material changes; protect the main branch once initial scaffolding is stable.